The Early History of Chinese Language

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Chapter 1

The prehistory of the Sinitic family

In this section, we trace the earliest stages of the Sinitic family, namely the pre-Classical stage, and discuss how it bridges the remote ancestor and the Classical period.

1.1 The Trans-Himalayan family

The Sinitic language family, often collectively referred to as Chinese, is a branch in the Sino-Tibetan language family, which is also known as the Trans-Himalayan language family. While the internal structure of the Trans-Himalayan language family and consequently a reconstruction of Proto-Sino-Tibetan with broad consensus (at the level of e.g. Proto-Indo-European) remain to be explored, the shared origin of Chinese, Tibetan, Burmese, and multiple other language families however is beyond a reasonable doubt: for example, a list of cognates and a tentative Trans-Himalayan reconstruction can be found in Hill (2019).

Having established a "flat" relation between Chinese and other languages in the Trans-Himalayan area, we may be interested in the position of Chinese within Trans-Himalayan, and in particular, when it diverged from the rest of the language family. Note that this is to be decided by comparison of cognates, and *not* by typological features (§ 1.2). This is inevitably related to the internal structure of the language family. A consensual phylogeny based on the comparative method is still lacking. Some scholars take a rather pessimistic view and suggest that it is probably a better idea to remain agnostic about higher-order subgrouping and adopt a *Fallen leaves model*, i.e. a thorough enumeration of all genetically certain subgroups without any clear genetic relations stipulated above them.

Two Bayesian phylogenetic studies (Sagart et al. 2019; Zhang et al. 2019) both consider the Sinitic branch to be the earliest branch that diverged from the rest of the language family, thus supporting the traditional binary division between the Sinitic branch and the Tibeto-Burmese branch.[1] They however cannot replace human readable and interpretable comparative analyses, and currently, the name *Tibeto-Burmese*, strictly speaking, is still not guaranteed to be a node in the phylogenetic tree of the Trans-Himalayan family, but rather a catch-all term for everything non-Sinitic.

[1] It should be noted that the huge divergence of contemporary Chinese languages from their Tibeto-Burmese cousins cannot be just explained by Neogrammarian sound change laws: language contact is an important factor.

1.2 On "Sino-Tibetan typology"

1.2.1 Purported typological similarities and counterexamples

Besides cognates, linguists occasionally talk about "Sino-Tibetan typology". Typological features often attributed to Sino-Tibetan languages include the isolating grammar, tonality, and monosyllabic roots.[2] These criteria was even once used to demarcate the boundary of the Sino-Tibetan language family, which later proved to be not reasonable. A refutation on the phonological part can be found in Matisoff (1973): features like tones and monosyllabic roots are not diachronically stable at all. As for morphological complexity, just think about how drastically the English inflectional morphology has simplified.

Recent linguistic fieldwork has also revealed that several previously underdocumented Trans-Himalayan languages completely violate these generalizations. The Gyalrongic languages, for example, have rather complicated inflectional morphology and therefore are definitely not monosyllabic, and Japhug, a Gyalrongic language, lacks tonal contrasts (Jacques 2021).

1.2.2 Typological classification of Proto-Trans-Himalayan

Given the striking internal typological diversity of Trans-Himalayan languages, the typological status of Proto-Trans-Himalayan is problematic.

1.2.2.1 Polypersonal indexation

We have reasons to suspect that Proto-Trans-Himalayan had a polypersonal indexation system similar to that of Rgyalrongic, actually. We note that the Rgyalrongic family and the Kiranti family are two distant groups within the Trans-Himalayan family (which is expected given that they are separated by the Himalayas) with seemingly cognate personal agreement affixes which is hard to be explained solely in terms of grammaticalization of pronouns, and one possible trace of personal agreement is also found in Tibetan (Jacques 2012; Jacques 2010; Jacques 2016). Considering the fact that the number of potential cognates between Rgyalrongic and Kiranti is around 150, and the number of cognates of them with Chinese is of the same order of magnitude,[3] we may assume that a Rgyalrongic-like personal indexation system originated at a quite early stage, which possibly is also the ancestor of Chinese (Jacques 2012).

Now, one possible objection comes from the fact that what exactly can be reconstructed to Proto-Sino-Tibetan is not without controversy. (a) We may assume that we have a Rgyalrongic-like polypersonal argument indexation system, which is a direct-inverse system, in Proto-Sino-Tibetan. (b) Or we can always argues that what can be reconstructed to Proto-Sino-Tibetan for certain is just personal affixes, as the underlying structure of a direct-inverse system is often not *that* different from more

[2] Modern Sinitic languages are by no means monosyllabic, while Classical Chinese appears to be monosyllabic: and disyllable words in later stages of Chinese seem to be due to either borrowing or syntactic fossilization of previous analyzable phrases. On the other hand, pre-Classical Old Chinese is possibly not strictly monosyllabic according to recent reconstructions (§ 3.2.1).

[3] See Shuya, Jacques, and Yunfan (2019) for an explicit historical comparison between Rgyalrongic and Sinitic.

common alignments (Oxford 2017), and therefore the direct-inverse system itself possibly cannot be dated back to Proto-Sino-Tibetan and only the personal affixes can, which are the diachronic materials used to form a Rgyalrongic direct-inverse system in later stages. (c) As a step further, it can be argued that the personal agreement affixes were probably of pronominal origins, and hence we can argue for a morphologically impoverished Proto-Sino-Tibetan, and thus, the great typological distance between Proto-Sino-Tibetan and Chinese is no longer that striking.

The viewpoints listed above are critically examined in Jacques (2012). Although a comprehensive comparison is still lacking, Jacques notices that there exist agreement affixes in Rgyalrongic and Kiranti that appear to be cognates and yet cannot be easily explained by grammaticalization of pronouns and hence the assumption (c) cannot be completely right, thus at least implying certain verbal morphological complexity in a common ancestor. Further, by comparative analyses, he demonstrates that the verbal conjugation paradigms in Rgyalrongic and in Kiranti are remarkably similar (for example, only transitive verbs come with the aforementioned affixes), hinting to a biactantial agreement system in the common ancestor, and that Kiranti seems to retain certain residues of an inverse prefix, while the evolution of a direct-inverse system, despite theoretically possible, remains to be rare in Eurasia. The statement (b) above therefore is also not quite feasible, although not completely impossible. Thus, a Proto-Trans-Himalayan with rich verbal morphology can be reasonably assumed.

One may then object by saying that that Rgyalrongic and Kiranti languages may have close genetic relations, and the complex verbal morphology is a shared innovation of that subbranch (which has been named as Rung languages). Jacques and Pellard (2021) examine this shared innovation hypothesis and list several problems of the hypothesis. First, possible reflexes of the purported Proto-Sino-Tibetan verbal system can be found in languages beyond Rgyalrongic and Kiranti, and the languages manifesting these features, i.e. the "Rung" languages, are geologically distant from each other. The shared innovation hypothesis has to come with a convincing theory of migration. Second, the "Rung" languages with the shared verb morphological features do not appear to have any unique shared lexical innovations: instead, forms with etymological relations can be easily established between "Rung" languages and non-"Rung" languages. Third, based on lexical evidence, internal classifications of the Trans-Himalayan family that are in conflict with the Rung hypothesis have been proposed. Thus the shared verbal morphological formatives found in "Rung" languages should not be taken as a shared innovation. Rather, retention of an older system is a better explanation.

1.2.2.2 The "causative" prefix

Another morphological device that probably has a Proto-Trans-Himalayan origin is the the **s-* prefix, which is also supported by reconstruction of Old Chinese. It sometimes is known as the causative prefix, although this usage is possibly due to grammaticalization of denominal verbs (Jacques 2015).

Chapter 2

Old Chinese philology

Middle Chinese phonology was already largely comparable to contemporary Sinitic languages. Having established that Proto-Trans-Himalayan is morphologically rich, the evolution from it to today's largely analytic Sinitic languages is rather intriguing. There has to be a series of transitional stages between an ancestor language with somehow Gyalrong-like typological traits and the modern Sinitic family, and a successful reconstruction of Old Chinese also helps understand these stages.

The Old Chinese period is quite long, spanning more than 1,000 years. Conventionally, the term *Old Chinese reconstruction* is mainly about the reconstruction of the language in the *Pre*-Classical period. Thus, we make use of Pre-Classical texts, including texts from Shang and Zhou dynasties (§ 2.1, § 2.2) and also evidence from neighboring languages, including early loan words in Old Vietnamese and Japanese. Finally, modern 闽 Min languages have features that cannot be explained by the Middle Chinese phonology (see below), meaning that there exists a stratum in Min languages which was a sibling, and not a daughter, of Middle Chinese. This non-Middle Chinese Sinitic branch is often known as Proto-Min. Finally, Middle Chinese rhyme books and rhyme tables provide a base line of whatever reconstruction we can think of, as we expect that the reconstructed Pre-Classical Chinese forms should explain what we observe in Middle Chinese.

An overview of available materials can be found in Baxter and Sagart (2014, Ch. 2). Here we introduce the more conventional sources of Old Chinese, namely Pre-Classical texts.

2.1 Oracle bones

The earliest attested Sinitic texts were oracle bone inscriptions from the Shang dynasty. It should be noted that people during the Shang dynasty *wrote* (and not just *inscribe*) as well: the counterpart of character 册 exists in oracle bones as well, which looks a bundle of bamboo and wooden strips. Still, the only materials that reflects the language of Shang currently available are inscriptions.

The characters in these inscriptions are structured in a way largely comparable to later Chinese characters. Since the phonetic values of characters in oracle bone inscriptions are not self-evident, some may even question whether these inscriptions are Sinitic at all: some may argue that the language of Shang was a (possibly unclassified) SVO language, and when the Zhou dynasty was established on the ruins of Shang, its culture was borrowed, including both the writing system and the SVO syntax. This

hypothesis is not quite plausible, because in such a scenario, usually what happens is *lexical* borrowing, and the introduction of a writing system should be minimal to the structure of the language (DeLancey 2013, p. 88). The introduction of Chinese characters to write Japanese is a good reference point. Further, phonetic borrowing (假借, 'borrowing, i.e. a character originally representing one morpheme to represent another morpheme with a close sound') in oracle bone inscriptions neatly fits into reconstructed Old Chinese phonology (e.g. Baxter and Sagart 2014, pp. 115).

2.2 Post-Shang texts

Oracle bone inscriptions are a 20th century archeological re-discovery not known to Classical Chinese authors. Zhou's replacement of Shang seemed to be accompanied by a great cultural revolution: the last king of Shang, namely King Zhou of Shang, was known by his arrogance, his corruption, and his tyrannic reign. Yet no one seems to remember the arguably most scary aspect of Shang dynasty: human sacrifice. To Classical authors, the earliest available texts are documents preserved in《尚书》(lit. 'venerated documents'), often known as the *Book of Documents* in English. Since these texts are from ancient kings whose deeds were romanticized by Confucian scholars, these texts were highly venerated and yet deemed as 诘屈聱牙 ('twisted, hard to pronounce') by post-Classical authors.[1] They were something that had to be read with commentaries, the latter written in easier Classical Chinese. These documents therefore should be regarded as pre-Classical, although they did contribute sporadic phrases and grammatical words (e.g. the copula 惟 or the pronoun 厥) that were occasionally used in Classical Chinese works as a way to polish an article.

A linguistic trait, which corroborates the cultural paradigm shift after the end of the Shang dynasty, is that the language of the *Book of Documents* and the language of oracle bone inscriptions are clearly different. The most notable fact on this aspect is that the aforementioned pronoun 厥 appears frequently in the *Documents*, but it appears neither in oracle bone inscriptions nor in Spring and Autumn works. Presumably, *Book of Documents* contains predominantly early Zhou dynasty texts, while oracle bones dates back to Shang, and the differences we are observing reflect dialectal differences between the ruling classes of the two dynasties.

Another fairly early source is《诗经》(lit. 'poem classics'), also known as the *Book of Odes*, which contains poems dates back to as early as early Zhou. The *Odes*, despite often considered, classical especially in a Confucianism context, differs greatly from Classical *proses*. This work mostly concentrate on the grammar of proses.

[1] See e.g.《进学解》(*Analysis of academic advancement*) by Han Yu.

Chapter 3

Outline of Pre-Classical Old Chinese

3.1 Methological issues

None of the texts listed above is alphabetic, and phonological information is extracted from the texts by observing (a) information contained in the phonetic components of Chinese characters (in transmitted texts, and also in excavated texts), (b) the rhyme patterns in the *Book of Odes*. (a) is done by observing 諧聲 *xiéshēng* 'harmonic sound' relations, i.e. semi-homophonous relations observed in Chinese characters without semantic relations that share a component. Based on inspecting characters with shared components but unrelated meanings in Modern Sinitic languages, we conclude that what is shared is likely a component representing the *initials* of the words they represent.[1]

Due to the uncertainty surrounding the exact phonological information hidden behind semi-homophonous character series and rhymes in *Odes*, as well as the uncertainty of the meaning of certain terms used in ancient works on Middle Chinese (e.g. Box 3.1), Old Chinese reconstruction is expectedly a chaotic field with lots of controversies. Baxter and Sagart (2014) represents an important milestone in this field, making use of all the materials mentioned above as well as a rich toolbox of philology, the historical comparative method, and internal reconstruction. But even it is met with some negative reviews (Harbsmeier 2016). Here we list some major methodological problems and not technical errors.

First, currently we do not even have a well-defined target of reconstruction. Pre-Classical texts show wide variation in syntax and presumably in phonology, as is shown in § 2.1 and § 2.2, and the Chinese scholar tradition clearly states that *Book of Odes* consists of poems collected at different times and different locations (Harbsmeier 2016, pp. 478-480). In the Classical period, texts made it clear that mutually incomprehensible Chinese varieties existed (Harbsmeier 2016, pp. 446-447). It is not clear if contrary to the claim of Baxter and Sagart (2014, p. 2), a distinction still needs to be made between Old Chinese(s) and Proto-Sinitic.

Second, even when the reconstruction is based on texts known to be from a relatively well-defined *branch* of Old Chinese, based on e.g. consistent morphosyntactic features, there is no guarantee that it has a single, unitary phonology (Harbsmeier 2016, pp. 488-490). Note that the same problem occurs for Middle Chinese as well (Box 3.1)

[1] In modern Sinitic languages, a character usually represents a root, but simple philological observation reveals that Old Chinese words were mostly represented by single characters.

Box 3.1: How unitary is Middle Chinese?

It should be noted that the problem is not restricted to Old Chinese. Strictly speaking, that there was a unitary Middle Chinese as reflected by rhyme books and rhyme tables also should not be taken for granted. What rhyme books and rhyme tables were intended for could probably never known for certain: they could be prescriptive works documenting the mixture of several respected dialects, or even an early Neogrammarian-ish reconstruction attempt. Besides, we are also not really sure about the phylogenetics of modern dialects: if we are to stipulate a Middle Chinese *node* in the Sinitic family tree which is the common ancestor of all modern Sinitic dialects, then how did it emerge from the messy congeries of dialects during the Middle Chinese *period* (Harbsmeier 2016, pp. 477-478)? We are more confident when talking about a unitary Middle Chinese (defined in the rhyme books) as the common ancestor of modern Sinitic languages, mainly because comparative works do suggest that the former is indeed a good starting point to analyze the phonology of the latter. Thus, Middle Chinese in the rhyme books at least is a good proto-language in the Neogrammarian sense. Still, Neogrammarian comparative studies based on modern Sinitic languages are urgently needed, not mentioning the fact that many are quickly going extinct without proper documentation.

Third, what exactly the rhymes and the *xiesheng* relations mean remain not completely clear. Rhymes can be cultural, containing traditional strata that no longer reflect the actual phonology when the poem was being composed (Harbsmeier 2016, pp. 454-455). The same problem exists for arguments based on shared phonetic components of Chinese characters. Some, for instance, argue that certain *xiesheng* relations observed that immediately lead to reconstruction of consonant clusters (§ 3.3.2) may simply be due to two pronunciations existing for one character, which was used to represent two initials because of that.

The severity of this issue however is more controlled than the severity of the first two morphological issues, namely the lack of language uniformity in the Pre-Classical Old Chinese period. First, there is no evidence in the earliest texts that suggests an earlier poem tradition, and the risk of rhyming being largely a cultural artifact is smaller than, say, rhyming in English literature. Second, it is always possible to base the reconstruction of syllable finals primarily on internal reconstruction of Middle Chinese, with *xiesheng* and *Odes* evidence being used primarily in fine-tuning (Hill 2019, p. 188).

Due to all of the issues and difficulties, we will avoid presenting a fine-grained reconstruction of Old Chinese (Classical or pre-Classical) phonology here. Below, we list some widely accepted features of (pre-Classical) Old Chinese to give the reader an intuitive feeling of how it feels like as a language. This may provide some insights for understanding the origin of certain features of Classical Chinese, like how valency alternation works. A full survey on the progress of Old Chinese reconstruction is beyond the scope of this work.

3.2 Morphosyntax

A trivially true fact often ignored is that what is transmitted in language evolution is words, either phonological words with grammatical significance, which may be a grammatical word or a short phrase that possibly undergoes reanalysis (as in some English dialects, *an uncle* > *a nuncle*), or a long grammatical word which may have multiple phonological domains. Therefore, a discussion on Old Chinese reconstruction

inevitably involves speculations on its morphosyntax.

Due to the nature of available Old Chinese materials, the more syntactic aspects of Old Chinese are much more easily known. Texts during the Old Chinese period have a grammar that at least look analytic, contrary to, say, Rgyalrongic languages. The evolution towards a largely analytic syntax therefore had to already have started at the Old Chinese stage.

The grammar of Classical and pre-Classical texts is on its own a enormous field, beyond the scope of this work. We just point out that the morphology had likely simplified considerably, such that there is almost no device in traditional texts that transparently records affixation. However, (residues of) morphology can still be observed, as well as deviations from the overall monosyllabic scheme of word structure. The rest of this section is devoted to its analysis.

3.2.1 "Bisyllabic" words

The format of the *Odes* strongly suggests that the language(s) behind the poems had largely monosyllabic words. This, together with the absence of explicit marking of affixation in the texts, means that the Old Chinese word structure is largely identical to its syllable structure.

One quick correction needs to be made to this picture. We have evidence for residues of minor syllables before the main syllables, which count as a syllable in the *Odes*. In *Odes*, some rather strange forms, like 帝命不時, contain 不 that would otherwise be interpreted as a negator, but the interpretations of them contain no negative meanings according to commentaries. Note that not all of them can be naturally explained as rhetorical questions.

We can also find records like 聿……吳謂之不律 in 说文, further suggesting the existence of such minor syllables (Hill 2019, pp. 122-1235). The reconstruction of their exact forms is still controversial, but their existence at least in some historical stages and areas seems to be beyond reasonable doubts.

These philological observations are considered as "the strongest motivation for faith" by Hill (2019, pp. 122-124).

3.2.2 Reduplication

By regular reduplication, disyllabic words can be created in Old Chinese. The available derivational devices that involve reduplication are in fact quite rich (Sun 2008).

3.2.3 Suffixes

The fact that Chinese tones probably originated from word-final consonants and the fact that in reading traditions of Classical texts, tone change represents subtle modification of the meaning of a root (known as 四聲別义 'four tones distinguishing between meanings') immediately suggest that Old Chinese has derivational suffixes.

All derivations that change the tone of a word to the DEPARTING sound, for instance, can now be explained by a **-s* suffix.

3.2.4 Prefixes

Several prefixes have proposed as well, based on alternations of the onset, e.g. multiple **s-* prefixes (Baxter and Sagart 2014, pp. 53-59).

3.2.5 Word structure, root structure, and phonotactics

To summarize, in Old Chinese, a word consists of a root with possible affixation, and is usually monosyllabic but can have a precedent minor syllable, and can also be disyllabic due to reduplication.

The question remaining to be asked is the phonotactics of the root and the word. For instance, one may wonder if the root is unable to have a consonant cluster onset, or whether the word (with all affixes included) to have a consonant cluster onset, or coda.

Baxter and Sagart (2014, pp. 50-52) proposes that the canonical root structure of Old Chinese can be roughly represented as Fig. 3.1.

Figure 3.1: Root structure

(C(ə)) C (r) V C (ʔ)

We have seen that the possible minor syllable has been included in Fig. 3.1, known as *loose pre-initials* in Baxter and Sagart (2014). What is particularly worth noticing is that they also allow *tight* pre-initials, without the **ə* vowel. An obvious reason to postulate tight pre-initials is that we have evidence to believe that Old Chinese allowed consonant clusters in the onsets of words, with philological evidence, especially characters with *xiesheng* relations which however represent words with radically different initials even in Middle Chinese (§ 3.3.2).

What remains to be known is whether this needs to be reconstructed as a part of the *root* phonotactics, that is to say, whether these consonant clusters are *always* results of prefixing.

To this question Baxter and Sagart (2014, p. 51) straightforwardly admit that they have no definite answer. Quite a few segments that look like pre-initials of roots are in fact prefixes, but it is hard to reduce all of them to prefixing. They thus place a period between a pre-initial that may or may not be a prefix and the main part of the syllable (1). Similarly, some loose pre-initials may also be prefixes but we are not sure, and a period is also placed between them and the main syllable of the root.

Note that here we have implicitly changed the meaning of the term *pre-initial*, and use it to refer to the first part of a *word*, and not just a root. This however is not evitable when derivation in Old Chinese has not been thoroughly understood, and the distinction between roots and words remains opaque in some cases.

(1) 椎 **k.druj* (Baxter and Sagart 2014, p. 279)

What Baxter and Sagart (2014) does not make clear is whether phonotactics laws similar to Fig. 3.1 can be reconstructed for the phonological structures of words. They certainly have no problem reconstructing forms with multiple prefixes and hence a remarkably complex initial consonant complex, based on converging evidence from Middle Chinese initial alternation of words with connected meanings and thus likely having the same root and different affixes (2).

(2) Baxter and Sagart 2014, p. 59

a. 登 *tˁəŋ > *tong* > *dēng* ‘ascend’ (bare root)

b. 增 *s-tˁəŋ > *tsong* > *zēng* ‘to increase (v.t.)’ (< ‘cause to ascend’; causative prefix **s-*)

c. 層 *N-s-tˁəŋ > *dzong* > *céng* ‘doubled, piled up’ (stative derived by prefixing **N-* to the causative)

For these forms, the term *pre-initial* is not necessarily well-defined, because it is currently not theoretically impossible to have a loose *prefix* (allowed by Baxter and Sagart 2014, p. 54) before a root with a loose pre-initial. Such forms do not appear in existing reconstructions, perhaps because of a *word*-level phonotactics law. But this has not been made explicit by any historical linguist. In a mature reconstruction of Old Chinese, we need to have both root-level phonotactics and word-level phonotactics, and sound laws targeting words predicting the Middle Chinese reflexes need to be based on word-level phonotactics if they make use of concepts like *pre-initial*.

Baxter and Sagart (2014) do have at least one *word*-level phonotactics law. They propose that the medial **-r-* in a word is either from the root or an infix (p. 51). This, together with the possible law that affixation does not create double minor syllables which we take as a working hypothesis and which means the pre-initial is always well defined for a word and not just its root, means that the phonotactics of Old Chinese words is not dramatically different from that of roots, except that the former allow more complex initial consonant clusters.

One natural question is whether there is any systematic relation between loose and tight pre-initials (understood as the first part of words). This question is especially concerning when it comes to affixation, because most loose pre-initials are assumed to be dropped in evolution towards Middle Chinese, and therefore if a prefix is assumed to have both a loose version and a tight version (Baxter and Sagart 2014, p. 54), then essentially one has a free parameter to arbitrarily explain why residues of the prefix appear in some Middle Chinese forms but not others, which obviously undermines the methodological principle of exceptionlessness of sound laws (Hill 2019, p. 126).

Needless to say, although each part of the grammar sketch above is supported by philological evidence, whether they are true (and if they are true, their precise forms, like 1 and 2) can only be told by comparing the reconstructed Old Chinese with both Middle Chinese and other Trans-Himalayan languages and checking if regular sound laws can be established between them.

3.3 Phonology

3.3.1 Tones and syllable finals

In Middle Chinese rhyme books, there are four tones reported: the LEVEL tone (平聲), the RISING tone (上聲),[2] the DEPARTING tone (去聲), and the ENTERING tone (入聲).

[2]The term may be confusing for a Modern Standard Mandarin speaker, as its reflex in Modern Standard Mandarin is the third tone, while the “raising” tone in Modern Standard Mandarin is the 陽平 tone, which historically was from the LEVEL tone.

3.3.1.1 The ENTERING tone

Let us first focus on the ENTERING tone. It can be easily seen that rhymes falling under the category of the ENTERING tone are much fewer than rhymes pertaining to the other three tones, with a lot of nasal rhymes missing. The reflex of roots with ENTERING tone in modern Sinitic languages, if any, all end with a stop (consider e.g. Cantonese *-p*, *-t*, *-k*). These words, when borrowed into Japanese, usually became disyllabic. These facts can be explained perfectly by assuming that in Middle Chinese, they ended in *-p*, *-t*, or *-k*, and this seems to be the defining feature of the ENTERING tone, which, in Middle Chinese, was strictly speaking not a tone at all.[3]

Now given that modern Sinitic tones can be easily explained by conditional sound changes starting from three tones (Sagart 1999a), the rest three tones in Middle Chinese were real tones. But we can try to apply the methodology inspired by the analysis above to Old Chinese, and the conclusion is that there was probably no tone in the Old Chinese period. Below, I briefly repeat the argumentation in Sagart (1999a).

3.3.1.2 The DEPARTING tone

The origin of Vietnamese tones has been demonstrated to be final consonants. That in the oldest layer of Chinese loan words into Vietnamese, words in the Middle Chinese DEPARTING tone regularly correspond to Vietnamese words in a tone originating from the final consonant *-s* strongly suggests that in Old Chinese, the origin of the DEPARTING tone is also something like **-s*. This is supported by textual evidence: in early Chinese transcriptions of foreign words, the DEPARTING tone sometimes corresponds to foreign *s*, and some early Chinese load words into Korean even retain the *s*. Therefore, the DEPARTING tone is likely a reflex of the Old Chinese word ending consonant **-s*.

3.3.1.3 The RISING tone

In a similar way, the RISING tone is related to an earlier glottal stop based on evidence from Sino-Vietnamese words. We also note that RISING tone words were preferred to transcribe Sanskrit short vowels, which is consistent with the glottal stop assumption, as the latter makes the syllable sound short. Thus, the origin of the RISING tone can be tentatively reconstructed as **-ʔ*.

3.3.1.4 The LEVEL tone

Now that all other tones have been reduced to syllable final consonants, the remaining LEVEL tone can be simply understood as the "default" case, i.e. open syllables or syllables ending with nasals. We have thus shown that Old Chinese lacks tones.

3.3.2 Initial consonant clusters

The syllable structure of Modern Standard Mandarin and other modern Sinitic languages is relatively simple. The syllable structure of Middle Chinese was slightly

[3] In modern Cantonese there are actually three reflexes of the ENTERING tone and they are real tones, but these tones are absent in ancient scholars' analyses of Middle Chinese, and are likely recent innovations.

more complicated, allowing more syllable-final consonants (i.e. ENTERING tone discussed above). However, all existing materials point to a much more complicated Old Chinese syllable structure.

A popular perspective is that consonant clusters were prevalent in Old Chinese. This perspective is not universally accepted, but alternative theories often either are not predicative or essentially only push the problem to an earlier stage.

3.3.2.1 Evidence from *xiesheng* series

One evidence for existence of initial consonant clusters in the Old Chinese syllable structure is *xiesheng* series with rather different Middle Chinese pronunciations.

Consider the semi-homophonous characters 监, 槛 (pronounced as *jiàn* in the reading convention of Classical texts). Then consider another series 滥, 蓝. The fact that they share the same phonetic component 监 strongly suggests that the four characters belong to the *same* series when the characters were first invented. A reconstructed initial like **kl-* therefore may be proposed for the phonetic value of the phonetic component 监. A similar "broken" series include 禀, 凛 (something like **pl-*). And now we find a class of consonant cluster initials roughly with the structure of **Cl-*, which actually is reconstructed by Baxter and Sagart (2014, pp. 50-51) as **Cr-*.

To deny that this phenomenon can be used to support consonant clusters in Old Chinese syllable initials, one may postulate that one character already had multiple pronunciations in the era of the *Odes* or even oracle bones, and thus a single phonetic component may be in two completely different *xiesheng* series. Now, proponents of such a theory then should ideally be able to explain the origin of these multiple pronunciations. If the explanation is still "they were once similar" then such a theory only pushes the problem to an earlier stage, and the reconstruction methodology above then applies to that stage, and at least we are not wrong when saying that consonant clusters existed at a certain stage of Pre-Classical Old Chinese.

3.3.3 Final consonant clusters

Consonant clusters at syllable finals are also proposed.

3.3.3.1 Final consonant clusters from suffixing

Qing dynasty scholar 段玉裁 Duan Yucai noticed that in *Odes*, rhyme sequences in the LEVEL, RISING and ENTERING tones can be isolated (meaning that all rhymes in a poem have the same tone in Middle Chinese), while the rhyming of DEPARTING tone words appears to be random. His conclusion was that there was no DEPARTING tone in *Odes*.

Now we know that there was probably no tone at all at least in pre-Classical Old Chinese, and Duan's observation means that the final consonant **-s* corresponding to the DEPARTING tone was likely almost always not a part of the root, but something like a suffix. But if a word with the ENTERING tone (which means it ends with **-p*, **-t*, or **-k*) rhymed with a word that ends with **-s*, then the latter should also contain **-p*, **-t*, or **-k* near its end: and hence the **-ps*, **-ks*, **-ts* final consonant clusters (Hill 2019, p. 183).

3.3.3.2 Internal reconstruction based on Middle Chinese tones and rhymes

In Middle Chinese, there are four rhymes (祭泰夬废) in the DEPARTING tone that do not have counterparts in the LEVEL and RISING tones, and words with these rhymes are semi-homophonous to ENTERING words, based on *xiéshēng* relations seen in their characters. This means these rhymes probably ended with **-ps*, **-ts*, **-ks* in Old Chinese, and have both ENTERING and DEPARTING features. Consonant clusters appear here again.

We also note that there are RISING tone words with nasal finals in Middle Chinese: 上 itself is an instance. This suggests a consonant cluster roughly with the form of **-ŋʔ* at the end of the words, which probably should be **-ʔŋ* to make it looks typologically feasible.

We note that the form **-ʔŋ* can even explain the origin of the RISING tone, as it makes the pronunciation *weak* in the middle, which naturally is close to a pronunciation that is *low* in the middle, and hence the modern RISING tone.

Chapter 4

Language contact

To summarize what has been revealed in reconstruction works, absence of tones, words with minor syllables, and various affixation mechanisms were characteristics of Chinese at a certain stage (Sagart 1999b, p. 13). In the earliest Chinese texts, however, we already see indications of emergence of stereotypical Sinitic features: the syntax was already quite rigid, the number of available affixes was limited and their productivity was likely weakened, and the syllable structure was under simplification (TODO: ref). The factors pushing Chinese on its path to its appearance today are of interest.

4.1 Relation with Chinese characters

It is sometimes argued that the loss of morphology is due to the Chinese characters, which do not record inflection well. This claim is linguistically untenable, first because most people before modernity were illiterate, and second because it is quite straightforward to dedicate certain characters to represent inflection (which is indeed how the *kana* system used to write Japanese developed). A reasonable, but not consensually accepted hypothesis is that it was due to language contact.

4.2 Area typology

Certain languages in the Trans-Himalayan area that do not have clear genetic relations with the Trans-Himalayan family, on the other hand, do have the purported Trans-Himalayan typological features. Kra-Dai languages have these typological features as well (Sidwell and Jenny 2021, p. 434), and yet the most likely relatives of them are the Austronesian family, which do *not* have these typological features at all (Sagart 2004; Ostapirat 2005).

The evolution of tonal distinctions happens in Kra-Dai as well (Sagart 2019).

4.2.1 Neighboring languages

Certain records in the Classical period indicate the existence of non-Sinitic populations in southern kingdoms. For example, the 越人歌 *Yuèréngē* '*Song of Boatmen of Yue*', a song of boatmen preserved in *Garden of Stories* that was both transcribed in Chinese characters (in a sort of Old Chinese pronunciation) and translated, was proposed to be composed in a Kra-Dai language (Zhengzhang 1991). What we are discussing above

happened long *before* the Classical period, but the relevant populations did not come out of no where and should be there as well when Proto-Sinitic was being formed.

Major non-Sino-Tibetan language families in touch with include Kra-Dai, Hmong–Mien, and Vietic. Among them,

What is particularly intriguing is that around 20% of words in McCraw (2010) do not have a clear origin. These words are often collectively referred to as having *Dongyi* 東夷 'eastern babarians' origins in modern Chinese scholarship, because these words have to come from language contact, and the most likely candidate of the origin is these "barbarians" i.e. non-Sinitic populations in the west of Shang and Zhou. Therefore, it seems that a ghost population once existed between the northern Sinitic population and the southern Austronesian population, speaking a language that heavily influenced Proto-Sinitic. No known surviving descendant of Dongyi is known.

4.3 The makeup of the Old Chinese lexicon

Less than a half of Old Chinese words have clear etymological Tibeto-Burman counterparts, and the rest of the lexicon comes from Austro-Asiatic and Tai contributions, with the former being the majority (McCraw 2010). Whether this means Old Chinese was more heavily influenced by Austro-Asiatic than other Trans-Himalayan languages is unclear, as a comparison between Austro-Asiatic loan words in Trans-Himalayan is currently lacking. Also worth noticing is that although in later periods, Chinese (at different stages of development) was the *origin* of borrowed terms, in the prehistoric era, Chinese was frequently the *recipient*, not the *donor*.

4.4 Syntax of Old Chinese and language contact

A natural idea is that the syntax of Old Chinese – and hence the syntax of Classical Chinese – was also influenced by non-Trans-Himalayan languages. The SVO word order, the prepositions, and clause-initial subordinators all stand in contrast to Tibeto-Burman languages. On the other hand, Old Chinese has sentence-final particles, which seem to be more consistent with a SOV order, and occasionally we observe relics of a SOV order as in 不我知也; these features are not surprising within the Trans-Himalayan family.

How sentence-final particles were grammaticalized is not always clear: in oracle bone inscriptions there are already some SFPs; see http://ir.lib.scu.edu.tw:8080/bitstream/987654321/11005001.pdf: 句尾亦偶有少數的語氣詞

We note that the all of the Southeast Asian groups with SVO syntax seem to have this order as far back as we can trace, and therefore in the aspect of word order (and probably other aspects), Chinese is the *recipient*, not the *donor* (DeLancey 2013, p. 82-85).